

Chandra Mohan SS

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Summary: I am a seeker, passionate and curious about all aspects of the science, IT, technical, software, manufacturing, printing, functional coating, inks, material analysis and colorimetry, hue, chroma, saturation, brightness, digital camera. Like to think out of the box, proactive self-starter, have agile learning experience, energy, vision to fulfill the goal. I am adaptable and flexible, practical, tactical, strategic and creative, able to start as new from the beginning.

Education

Ph.D. Printing & Paper Science & Eng.(GPA 3.94),	Western Michigan University	Kalamazoo, MI, US (2019-2023)
Master's of Science in Print Media (GPA 3.84)	Rochester Institute of Technology	Rochester, NY, US (2017-2019)
Bachelor of Engineering in Printing Technology	BMSCE (VTU).	Bangalore, India
Diploma in Printing Technology	Board of Tech Education, GIPT.	Bangalore, India

Certification, Skills & Connections:

Print & Paper industry connections: TAGA, TAPPI, GAA, FTA, AIMCAL, FOGRA, ESKO, EFI, PGSF

Certification: Sun-microsystem (Oracle) SCJP-Certified Java Programmer, NIIT Bangalore, India

Skills: Printing technologies and processes, proofing, and color management, print applications; Esko Software Suite, JIRA and bug tracking software, MS Office, Agile Methodologies, Adobe Suite for Print, Customer Support, Training and Development, Presales and post-sales, Basics of Color Cert, Color Think Pro, ISO & Gracol2013. Equipment; Print media – Press and finishing, Basic Programming Languages and Database: JAVA, C++, HTML.Oracle9i

Awards & Recognition

Dusty Rhodes Graduate Student Paper Award, Technical Association of the Graphics Arts (TAGA), 2019

Technical Writing Contest Winner, Gravure Education Foundation. 2019, 2021.

FFTA Rossini North America Flexographic Research Scholarship 2021 First Place,

Printing United Alliance 2023-24 Scholarship Recipient.

Publications:

- 1) Sustainable and Eco-Friendly DLC Fabrication using PECVD for Replacing chrome on Gravure Cylinder Surface (Published to ACS Journal: Langmuir: <https://pubs.acs.org/doi/abs/10.1021/acs.langmuir.3c02177>)
- 2) DLC-engineered flat gravure surface: enabling sustainable fabrication to replace chrome for printing conductive line electrodes in flexible electronics (Published to RSC Journal Material Advances: <https://pubs.rsc.org/en/content/articlelanding/2024/ma/d4ma00562g>)
- 3) Durable Thin-Film DLC on Wafer Surfaces of Gravure Cylinders for Roll-to-Roll Printing of 1-Bit Electrodes and Microtext in Flexible Electronics and Graphic Security (Published to Wiley Material Advances: <https://doi.org/10.1002/adma.202413551>)
- 4) Srinivasaraju, C. S. (2019). A Comparison of Conventional and New Technology Gravure Cylinders in Terms of Image Quality (Masters thesis, Rochester Institute of Technology).

Professional Experience

Application Engineer (March 2024 – current)

Mondi Group (Ridgely/Leonia, NJ, US)

Focus Areas: Product Development, New Functional Surfaces for New Customers, Project Management, Material Cost Reduction, Quality Improvement, Material Analysis & Qualification. Vendor sourcing and negotiating for print materials.

- Development of release coating formulations, utilizing expertise in printing, coating, and drying techniques. Colorimetry and Color matching for customer approvals.
- Designed and executed experiments, inspection plans, and evaluation methods, leveraging SAP (Systems Applications Products) and MES (Manufacturing Execution System) for process optimization and data analysis.
- Supported the production and functional coating of a wide range of products, including rigid and flexible labels, tape release liners, siliconization, MG, MF, SCK, PCK papers, as well as PET and BOPP films used in packaging, hygiene, building & roofing, shingles, and battery liners.
- Conducted root cause analysis for customer claims, providing actionable insights to resolve issues related to paper, film, ink, and coating liquid performance.
- Managed relationships with paper, film, ink, and coating liquid suppliers to identify new alternative materials for coating.
- Collaborated with

production teams on statistical analysis methods to drive process improvements and meet production targets.

Skills: Analytical Tools: Hitachi XRF, Rigaku XRF, Oxford XRF, Coat Weight, Minitab, Tensile Test, Chem Instrument Release Test, Co-efficient of Friction Test, Tack Loss, SAP, New Product, Ink and Substrate, Coating and Ink Mixing.

Printing and Graphic Production Engineer Intern (Oct 2023 – Dec 2023)

IDEMIA (Exton/Philadelphia, PA, US)

Ink, coating & substrate qualification for payment card applications using 7 color Offset and silk screen printing process.

Skills: Silk Screen Printing, flood coating and Offset Print Manufacturing, Credit Card Outlay, Inlay and functional layer heat lamination, punching for singulation, metal card decoration, Art work, Ink and Substrate

Substrate and Print Professional Aid (Nov 2022 – May 2023)

School of Packaging Michigan State University (East Lansing, MI, US)

Design and perform experiments on different flexible packaging substrates relevant to the project for coating and printing. Develop and validate a conventional rotogravure and screen printing methodology for implementing active surfaces using coating methods for product packaging. Application of UV cure acrylics, epoxies, polyurethanes, and polyesters in the active surface coating.

Skills: Data Analytical Tools: FTIR, DSC, Surface Tension and Energy, Surface Treatment, SEM, UV Cure, Ink Draw Downs, Viscometer, Rheology, Cobb Test, Minitab, Electrical Resistance, Air Permeability, Water Vapor Transmission Test, Lab Safety and Quality Compliance

Digital Print Engineer (April 2022 – Sep 2022)

De La Rue Authentication (Minneapolis, Minnesota, US)

- This involved Inkjet Print Producer and print quality R&D initiatives. Collaborate with Senior Scientists and Engineers to develop high quality Inkjet print security label production machine that use Ricoh, Kyocera, Dimatix, Fuji StarFire, Xaar digital piezo crystal inkjet head technology
- Successful completion of Factory Acceptance Tests and Site Acceptance Tests for large scale digital press
- Extensive research on RGB to CMYK print process that emphasis the use of digital presses (inkjet heads and toner) in laboratory and production settings. Color Profile creation and Gamut analysis using X-Rite i1i0
- Operated Print Head Doctor, Kruss MSA Contact Angle, PIAS-II Image Quality, Motic Microscope, Barbieri Spectro LFP Calorimeter, Raster Image Processor: Color Gate Production v22 Raster Processor, and Elcometer Crosshatch
- Analysis of TDS and SDS of transparent and MET PET, PE, Polycarbonate, and Paper substrates to assess appropriateness for digital printing. Supplier and Vendor management
- Qualified ink and substrate, evaluating compatibility for print quality and longevity using Cross Hatch, Chemical resistance tests, measuring tools
- Tested ink performance to meet security printing prerequisites QR codes, Microtext, Fine lines, dots to fulfill certification requirements

Skills: Project Management, Security Print, Piezo Inkjet, Data Analytical Tools and Applications, Lab Safety and Quality Compliance

Teaching & Graduate Research Assistant (Jan 2020 – Dec 2021)

Western Michigan University (Kalamazoo, Michigan, US)

- Transformed the traditional in-person GPS 1500 Graphics and Printing Science, Fundamentals of Graphic Arts, Rotogravure and Color Management courses into online modules to facilitate learning for undergrad students during the COVID-19 pandemic, assisted in teaching. Served as a Graduate Teaching Assistant and offered guidance on lab and class coursework related to digital printing, prepress and press operations, screen printing, and Adobe Creative Suite, specifically Photoshop, Illustrator and InDesign applications for print.
- Instruments used: First Ten Armstrong Surface Tension Device, Anton Paar Rheology measurement devices, Bruker White-light interferometer, contact profilometry, reflectometry, confocal laser microscopy, optical microscopy and color measurement via spectrophotometry, X-Rite Spectrophotometer, and i1i0 for color profiling. Material Tensile Tester, FTIR, DSC.
- Supported paper manufacturing, coating, calendaring, and sizing to enhance print quality
- Subjects Studied as PhD Student: GPS 5201 Color Printing and Substrates, PAPR 5000 Introduction to Papermaking, PAPR 6410 Paper Coating Formulations, PAPR 5501 Advanced Paper Processes, GPS 5100 Printability Analysis, and PAPR 6950 Advanced Ink
- Pigment Ink and Coating formulation, Viscometer, Thinky Planetary Centrifugal mixer, for use in electronics and graphic experiment.

- Developed and implemented printing processes parameters for Flexography and Gravure to deposit or coating of Lithium Ion battery Anode inks on copper and aluminum substrates for use in coin cell energy devices.

Skills: Data Analytical Tools and Applications, Minitab, Lab Safety and Quality Compliance, WLI 3D Optical profilers.

Art & Print Producer, Communications Manager (May 2019 – Sep 2019)

Royal Caribbean International via DG3/Rayan Edwards (Barcelona, Spain)

- Demonstrated exceptional control on in-cruise print operations of the world's largest cruise ship, "Oasis of the Seas"
- Installed and maintained pilot Electrophotographic Konica Minolta 3780 Digital Machines, and executed pilot onboard design guidelines. Application of calorimeter and color profiling on electrophotographic machine.
- Coordinated with various departments to create a daily itinerary, maintain inventory, and order supplies of paper and toner, increasing machine uptime by 100%
- Managed shoreside collateral projects, and handled large volumes of digital assets for creative art printing and display management of onboard retail, spa and fitness graphics, art exhibits, signs, gift cards, and personalized premium items. Implemented incoming Art & Print campaign collateral initiatives, print work for OOH, DOOH, Display, Point Of Purchase (POP) for inside cruise shopping and special magazine for the cruise business.
- Fulfilled daily print production requirements for Cruise Captains Bridge, HR, and Engineering operations
- Designed and executed specialty and Atlantis European tour graphics and collaterals
- Reproduced over 15 million high-quality prints, managed 5 sign-in crews, and improved print operations quality with limited direction

Skills: Continuous Improvement, Stakeholder Communication, Collaboration, Team Leadership, Art & Print Production

Graduate Assistant (July 2017 – May 2019)

Rochester Institute of Technology (Rochester, NewYork)

- Investigated and Developed Test Methods to evaluate New Technology Print Cylinder image carriers and image quality for a European grant project. Application of Statistical Methods using Minitab for evaluation of results.
- Image Quality Analysis using Calorimeter, Betaflex Pro 3D and Dot Structure and fine line Analysis.
- Subject Studied: Color Physics, Color Management, Printing Materials and Processes, Package Printing.
- **Print Producer:** Mastered RGB to CMYK print process, operation and laser calibration of Kodak Approval Dye Sublimation to accurately proof images. Participated G7 Certification of Presstek 52DI waterless offset printing process, in print production of customer jobs. Overseen the operation of 10 stations, 8 colors automated screen-printing press with 54" IR dryer to produce over 15 thousand t-shirts in a month for RIT Sports, Ice Hockey, and Seneca Park Zoo
- Managed the operations of the Xerox 8080 and Xerox iGen 5 digital presses to create digital print enhancements.
- Supervised Finishing tasks such as Cutting, Folding, Booklet making, Perfect binding/bookbinding, Scoring/perforating/creasing, Mounting, Gluing, padding, inserting, and collating
- Monitored Epson Wide Format Printing, Heat transfer printing for full-color graphic printing on fabric and plastic.

Skills: Research, Lab Safety and Quality Compliance, Continuous Improvement, Stakeholder Communication and Collaboration, Project Management, Art and Print Production

Print and Software QA Engineer (June 2013 to Jan 2017)

Esko, a Danaher company (Bangalore, India)

- Part of a team responsible for package printing software development, including Imaging Engine, Color Engine, AE, and Proof.
- Software Test Case management for UI/UX, color, and proof workflows, Esko Color Pilot, profiling, calibration, and testing for Art & Print production communication and workflow
- Implemented and tested software test cases for package printing applications, Color Engine, Imaging Engine; ensured quality and ease of use of the software, reduced errors through personal review and team inspection, and trained internal employees on color applications

Skills: Continuous Improvement, Stakeholder Communication and Collaboration, Project Management, Team Leadership, Creative problem solving

Technical Associate IT (April 2011 - June 2013)

Dell International Services (DIS) (Bangalore, India)

- Mastered IT support software applications used in Travel & Leisure company TUI (Touristik Union International), including airways, retail stores, sales and marketing offices, cruises and resorts
- Provided expert remote troubleshooting and assistance for hardware and software issues for digital signage displays in over 3000 TUI retail stores across the UK and Europe.

- Technical support: Main frame server applications, Networking, Chip and Pin Card device transactions, operating systems, windows server computers, virtual networks, telecoms, and on-demand ticket printers for the travel and hospitality industry
 - Demonstrated exceptional print support service during peak holiday periods, aiding on-time art and print production for customer satisfaction, experience and promoting sales in the form of holiday packages
- Skills: Exceptional Learning, Continuous Improvement, Support Leadership, Creative problem solving, creative, production, and localization Technologically savvy

HP Inkjet Print-Technical Support Officer (March 2009 - March 2011)

Mphasis and The Hewlett-Packard Company (Bangalore, India)

- Demonstrated expert proficiency in HP Prometric exams for new print product. Highly influential in providing technical and image quality support for touch screen, wireless high-end HP Inkjet printers. Thermal Inkjet, Piezo Crystal Inkjet.
- Educated American and international customers on set-up, troubleshooting and printing using home and business inkjet printers for on-demand document and photo printing
- Mastered the hardware and software quality features of HP Single & Multicolor, All-in-one, Digital Inkjets, Scanner, Fax to help customer in resolving print quality issues and support product development team
- Achieved service level agreement of HP in satisfying the customer expectation and resolution

Skills: Continuous Improvement, Team Leadership, Technologically savvy

Pad Printing Technical Support Executive (2006 - 2007)

Kinnari Print Tech (Bangalore, India)

- Application of pad printing technologies for printing on Glass, Metal, Wood, Ceramics, Fiber and other irregular surfaces. Training and installation of Pad Printing Technologies at industrial customer locations.
- Responsible for pre-sale and post-sale technical support for pad printing technologies and business development.
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Junior technician quality assurance (2003)

Macmillan India Private Limited (Bangalore, India)

- Responsible for quality assurance, author supported editing and proofing of foreign scientific journals & articles such as Elsevier & Pergamon, John-Willy, Wiley-Blackwell UK, & the US. Duties include proofreading, checking of pagination, picture, and graphics as per the scientific journals' specifications.